

Name: _____

Solutions

Date: _____

Year 9 Mathematics**Test 8, 2015****Topic – Linear Equations**

35 = _____ %
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Total Time: 40 minutes**Total Reading:** 4 minutes**Total Working:** 36 minutes**SECTION 1: CALCULATOR FREE**

Time: 15 minutes	Marks for Section 1: 13
Reading: 2 minutes	Equipment Allowed: Nil
Working: 13 minutes	

1. [4 marks: 1, 1, 1, 1]**Write an equation for each of the following statements. Do NOT solve it.****a)** When I add five to a number I get twelve.

✓ $x + 5 = 12$

b) When I add three to five times the number I get fifteen.

✓ $5x + 3 = 15$

c) Three less than four times the number is nine.

✓ $4x - 3 = 9$

d) Number is divided by four and subtracted from fourteen. The result is negative eight.

✓ $14 - \frac{x}{4} = -8$

2. [6 marks: 1, 1, 2, 2]**Solve the following linear equations:****a)** $x + 7 = 23$

✓ $x = 16$

b) $\frac{x}{3} = 12$

✓ $x = 36$

c) $2x - 2 = 20$

$2x = 22$ ✓

$x = 11$ ✓

d) $\frac{3x}{5} = 6$

$3x = 30$ ✓

$x = 10$ ✓

3. [3 marks: 1, 1, 1]

✎ Complete the following sentences:

a) To solve an equation, we perform OPERATIONS in reverse order. ✓

b) A LINEAR
EQUATION has only one solution. ✓

c) The unknown number in the equation is called VARIABLE. (PRONUMERAL) ✓

~ END OF SECTION 1 ~

Name: _____

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SECTION 2: CALCULATOR ASSUMED

Time:	25 minutes	Marks for Section 2:	20
Reading:	2 minutes	Equipment Allowed:	$\frac{1}{2}$ page notes (A4 one side), CAS calculator
Working:	23 minutes		

4. [4 marks: 2, 2]

- a) A painter pours 4 litres of paint out of a bucket. This leaves 6 litres in the bucket. When full, the bucket holds x litres.

- i) Which of these equations could be used to work out the amount of paint in the bucket when full? *Circle*

A $4x = 6$

B $x+4=6$

C $x-4=6$

D $x+6=4$

- ii) Solve the equation to work out x .

$$x = 10$$

- b) A large number (x) of boys register for school rugby. The sports coordinator divides them up into teams of 15, and gets exactly 5 teams.

- i) Which of these equations could be used to work out the number of boys who registered?

A $5x = 15$

B $x+5=15$

C $x-5=15$

D $\frac{x}{15}=5$

- ii) Work out the number of boys registered.

$$x = 75$$

5. [12 marks: 2, 2, 2, 2, 3, 3]

Solve the following linear equations:

a) $6x = -20 + x$

$$5x = -20 \checkmark$$

$$x = -4 \checkmark$$

b) $\frac{x+6}{5} = 9$

$$x+6 = 45 \checkmark$$

$$x = 39 \checkmark$$

c) $\frac{x}{3} + \frac{1}{4} = \frac{1}{2}$

$$\frac{x}{3} = \frac{1}{2} \checkmark$$

$$x = \frac{3}{2} \checkmark$$

d) $5(x-3) = 2x$

$$5x - 15 = 2x \checkmark$$

$$3x = 15 \checkmark$$

$$x = 5 \checkmark$$

e) $5(x-12) = 2(x-3)$

$$5x - 60 = 2x - 6 \checkmark$$

$$3x = 54 \checkmark$$

$$x = 18 \checkmark$$

f) $\frac{11-2x}{4} + 3 = -6$

$$\frac{11-2x}{4} = -9 \checkmark$$

$$11-2x = -36 \checkmark$$

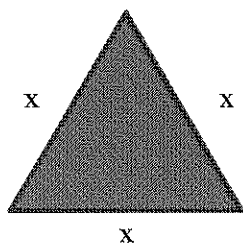
$$-2x = -47$$

$$x = \frac{47}{2} \checkmark$$

6. [4 marks: 2, 2]

Write an equation for each of the following statements then SOLVE it.

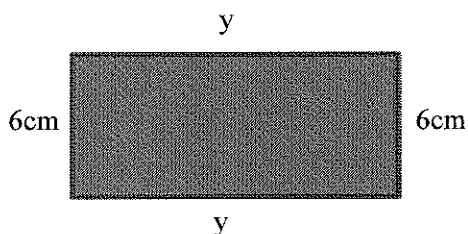
- a) An equilateral triangle has all three sides the same length. If the perimeter of an equilateral triangle is 21cm, what is the length of each side?



$$3x = 21 \checkmark$$

$$x = 7 \checkmark$$

- b) A rectangle is 6cm wide and its perimeter is 32cm. What is its length?



$$2y + 12 = 32 \checkmark$$

$$2y = 20$$

$$y = 10 \checkmark$$

~ END OF SECTION 2 ~